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A pan-cancer single cell landscape reveals heterogeneity and functional diversity of double-negative T cells

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Abstract

Background Double-Negative T (DNT) cells, lacking both CD4 and CD8 expression, play critical roles in cancer immunology, and have garnered increasing attention in cancer research. However, their heterogeneity and functional diversity within the tumor microenvironment (TME) remain underexplored.

Methods In-house and publicly available single-cell RNA sequencing (scRNA-seq) data for different cancer types were integrated after quality control and batch effect correction, followed by DNT cells separation from $CD3^+$ T cells subtypes. Functional characteristics, intercellular communication, differentiation trajectories, regulatory networks, and clinical relevance were analyzed among different DNT subsets. Key findings were validated using multiplex immunofluorescence and spatial transcriptomics to investigate the spatial localization of DNT subsets and their interactions within the TME. Impact of γδ T cells on immunotherapy response was also assessed using MC38-based murine tumor model.

Results By integrating scRNA-seq data from 2,369 samples across 23 cancer types, we established a comprehensive single-cell atlas of 157,025 high-quality DNT cells. Fourteen distinct DNT subsets (6 αβ DNT and 8 γδ T cell subsets) were identified, demonstrating tumor both type-specific and shared distribution patterns, as well as unique cell-cell interaction network within the TME. These subsets displayed specialized functional profiles, including cytotoxicity, antigen presentation, and immune modulation, indicating that the functional diversity of DNT cells is largely subset-

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specific rather than a manifestation of multifunctionality within a single population. We also delineated divergent trajectories for αβ DNT and γδ T cell subsets, including the functional plasticity of gut-resident γδ T cells transitioning between cytotoxic and immunosuppressive states. Notably, several DNT subsets were significantly associated with favorable clinical treatment outcomes, including improved responses to cancer immunotherapy. Consistently, depletion of γδ T cells in the murine tumor model significantly decreased the efficacy of PD-1 blockade, underscoring their critical role in therapeutic response.

Conclusions Our study uncovers the previously underappreciated heterogeneity and functional diversity of DNT cells in the TME and demonstrates their profound impact on tumor progression and immunotherapy outcomes.

Keywords Single-cell RNA sequencing, Pan-cancer, Double-Negative T cell, Γδ T cell, Tumor microenvironment, Immunotherapy, Heterogeneity, Functional diversity

Background

T cells, particularly $CD4^+$ and $CD8^+$ subsets, have been extensively studied over the past few decades, leading to a well-established understanding of their pivotal roles in anti-tumor immunity. However, double-negative T (DNT) cells, a subset that lacks both CD4 and CD8 expression, have recently emerged as important players in cancer immunity and remain relatively poorly understood [1]. DNT cells are categorized into αβ and γδ lineages based on their T cell receptor (TCR) and both of which originate from early thymic progenitors but follow distinct developmental trajectories [2]. Traditionally regarded as precursors to $CD4^+$ and $CD8^+$ T cells during αβ T cell development, recent evidence indicates that DNT cells in peripheral blood and tumor tissues are mature T lymphocytes, constituting a heterogeneous population with distinct phenotypes and functions [3]. DNT cells represent a small but consistent fraction of peripheral $CD3^+$ T cells, accounting for approximately 1–3% for αβ DNT cells [4] and 1–5% for γδ T cells [2]. Notably, γδ T cells preferentially reside in tissues (e.g., skin, intestine and lungs), where they contribute to localized immune responses and tissue homeostasis [5]. Clinical responses to allogeneic DNT therapies have been observed in both hematologic and solid tumors, suggesting their potential to overcome limitations associated with current $CD8^+$- or $CD4^+$-based immunotherapies, particularly in HLA class-I-deficient tumors [6, 7].

Although the importance and function roles of DNT cells in cancer immunity (e.g., cancer immunotherapy) have been recognized, much of the existing research relies on experimental approaches (e.g., flow cytometry), which primarily focus on mouse models or non-tumor contexts, leaving a substantial gap in understanding their roles in human tumor [8–10]. Recent findings have uncovered pleiotropic functions (e.g., cytotoxicity and antigen-presentation) of DNT cells [2, 11], raising the question of whether these diverse functions originate from a single, multipotent subset or from distinct, specialized subsets. Evidence increasingly supports the existence of diverse DNT with distinct functions, supporting their transcriptomic divergency in regulation, IFNγ-producing, granzyme-related cytotoxic, antigen-presentation etc [11–19]. However, TCR-based classifications of γδ T cells has proven to be inadequate for capturing their functional diversity and often yield inconsistent results across tumor types [2]. For instance, different γδ T cell subsets influence treatment outcomes in patients with cholangiocarcinoma [20], liver cancer, and lung cancer [21] compared to kidney cancer [22] and triple-negative breast cancer [23]. These discrepancies highlight the need for transcriptomic profiling to fully characterize DNT heterogeneity and functional diversity.

Recent advancements in single-cell RNA sequencing (scRNA-seq) have revolutionized oncology research by enabling the high-resolution analysis of the TME at the single-cell level [24]. However, efforts to investigate tumor infiltrating DNT cells using scRNA-seq have been limited by small samples sizes and insufficient cell numbers, restricting the ability to capture the full spectrum of DNT subsets and comprehensively characterize their phenotypes and functions. To obtain sufficient cell numbers, scRNA-seq has been performed following cell sorting to specifically characterize the DNT cells [25, 26], which unfortunately limits the samples size and restricts exploration of interactions between DNT cells and other TME components. On the other hand, integration of scRNA-seq datasets has systematically profiled major cellular populations (e.g., lymphoid [27–29], myeloid [30–33], stromal [34–38] and vascular components [39, 40]) across various cancer types, providing profound insights into cancer immunology and clinical applications at pan-cancer single-cell level [41]. As the increasing scRNA-seq dataset are available across different cancer types, it is possible to obtain enough tumor infiltrating DNT cells to investigate their heterogeneity and functional diversity.

In this study, we integrate pan-cancer cohorts to construct the most comprehensive single-cell atlas of human DNT cells to date, encompassing a wide range of tumor types. This approach enables robust matched comparisons and high-resolution analyses of αβ DNT and γδ T cells, identifying both previously reported and novel

DNT subsets. Additionally, we reveal the clinical significance of DNT subgroups, highlighting specific subtypes associated with favorable prognoses across multiple cancer types. These findings advance our understanding of DNT cell biology in cancer and underscore their potential for therapeutic development.

Methods

Single-cell library preparation and sequencing

The Chromium Single Cell 3′ Library & Gel Bead Kit v2 (PN120237), the Chromium Single Cell 3′ Chip Kit v2 (PN-120236), and the Chromium i7 Multiplex Kit (PN-120262) were utilized following the manufacturer's instructions outlined in the Chromium Single Cell 3′ Reagents Kits v2 User Guide. Prior to processing, the single-cell suspensions were washed twice with 1× phosphate-buffered saline (PBS) supplemented with 0.04% BSA. The cell count and concentration were verified using the TC20 Automated Cell Counter. Cells were then loaded onto a 10× Genomics Chromium Controller machine for gel bead-in-emulsion (GEM) formation. Barcoded complementary DNAs (cDNAs) were synthesized using the 10× Genomics Chromium Single Cell 3′ reagent kit (V2 chemistry), followed by recovery, purification, and amplification to yield adequate quantities for subsequent library preparation. The quality and concentration of the libraries were evaluated using the Agilent Bioanalyzer 2100, and sequencing was performed on the Illumina NovaSeq platform with PE150 sequencing.

Antibody-mediated depletion of intratumoral γδ T cells in MC38 tumor-bearing mice

MC38 cells (ENH204-FP, Kerafast) were cultured in DMEM supplemented with 10% fetal bovine serum (FBS) and 1% penicillin–streptomycin (PS) at 37 ℃ in a humidified 5% CO2 incubator. Female C57BL/6 mice (6–8 weeks old) were housed in a specific pathogen-free (SPF) animal facility. Mice were randomized into four groups ($n = 5$). On day 0, 1×10^6 MC38 cells were resuspended in 100 µl PBS and subcutaneously inoculated into the mice. Starting on day 2, mice were administered intraperitoneal injections of anti-mouse TCR γ/δ (BE0070, clone UC7-13D5, BioXcell, 300 µg/mouse) or an isotype control (Armenian hamster IgG, BE0091, BioXcell) for a total of five injections. From day 5 onward, anti-PD-1 monoclonal antibody (BE0146, clone RMP1-14, BioXcell, 200 µg/mouse) or an isotype control (BE0089, clone 2A3, BioXcell) were administered intraperitoneally every three days for a total of four doses. Tumor sizes were measured every two days using digital calipers. Subcutaneous tumors were collected on day 17.

Multiplex immunofluorescence

Multiplex immunofluorescence staining was performed using the Opal™ 7-Color Manual IHC Kit (Akoya Bioscience, NEL801001KT). Tumor tissues were obtained from West China Hospital of Sichuan University. Tissues were sliced into 5 µm sections and heat in the retrieval solution of citric acid or EDTA to recover antigens. All antibodies, CD3ε (85061, Cell Signaling Technology, 1:100,Opal 780), CD4 (ab133616, Abcam, 1:500, Opal 480), CD8α (85336, Cell Signaling Technology, 1:400, Opal 520), Anti-TCR beta (ab315324, Abcam, 1:400, Opal 690), Anti-human TCR γ/δ (331202, Biolegend, 1:100, Opal 570), Anti-Interferon gamma (ab231036, Abcam, 1:200, Opal 620), Anti-MHC Class II (ab55152, Abcam, 1:200, Opal 620), Anti-Integrin alpha 1 (ab243032, Abcam, 1:1000, Opal 620) were evaluated via immunohistochemistry. Briefly, the sections were dew-axed with xylene and graded alcohol. Antigen retrieval was performed in buffer (pH 9.0 or pH 6.0) for 20 min in Microwave at high power. Next, all sections were blocked using an antibody diluent/block (72424205, Akoya Bioscience) for 10 min, then incubated with a primary antibody for 1 h at room temperature or 4 C° overnight. Subsequently, sections were incubated with secondary reagents and tyramide signal amplification reagents for 10 min (Opal 480, Opal 520, Opal 570, Opal 620, Opal 690, and Opal 780. Akoya Bioscience, 1:200). MWT antigen retrieval was performed until all markers were stained. Finally, DAPI-based nuclear staining was performed at room temperature for 5 min. Then sections were covered with anti-Fade fluorescence mounting medium (ab104135, Abcam), and scanned by Vectra Polaris Automated Quantitative Pathology Imaging System. The data analyses were performed with QuPath software.

Spatial transcriptomics preparation and sequencing

Spatial transcriptomic sequencing was performed using the 10x Genomics Visium CytAssist platform. Each single formalin-fixed paraffin-embedded (FFPE) tumor tissue sample was obtained and meticulously mounted onto Visium CytAssist Gene Expression Slides. The tissue underwent a comprehensive series of preparation steps, including deparaffinization, staining, imaging, decrosslinking, and probe hybridization, to enable the accurate capture of spatially resolved mRNA transcripts. Following preparation, the gene expression slides were loaded into the Visium CytAssist instrument for bright-field imaging. Within the instrument, analyte transfer was executed to ensure precise acquisition of transcriptomic data from the FFPE sample. The bcl2fastq tools was used to convert raw base call files to FASTQ reads. The resulting raw sequencing data were subsequently processed with the Space Ranger pipeline (version 1.3.1) aligned to the human reference genome GRCh38-2020.

Single-cell data processing

The In-house scRNA-seqs data were processed following well-established analytical practices, using the same strategy as in our previously established workflow [34, 42–46]. Firstly, scRNA-seq data from 10X Genomics were aligned to the GRCh38 reference genome using the count application (Cell Ranger v6.0.2) with default parameters. Count matrices of cells by genes from Cell Ranger were used for subsequent downstream analyses. To ensure the capture of only high-quality cells, we filtered out potential doublets and low-quality or dying cells according to multiple quality metrics, including total UMI counts, the number of detected genes, and the fraction of mitochondrial gene expression per cell. Specifically, cells were removed: more than 10% mitochondrial gene expression, less than 200 genes expressed, and more than 4000 genes expressed. Cell doublets were identified using the *scanpy.external.pp.scrublet* function, with *expected_doublet_rate* = 0.05 from Scanpy package (version 1.9.8). Cells with doublet score greater than 0.25 were removed. For published available scRNA-seq data, we downloaded the 10x Genomics-derived count matrix (and raw FASTQ data if accessible) along with detailed metadata, including clinical information, demographic characteristics, and sample types. The same filtering steps were applied to the count matrix.

Due to the heterogeneity of scRNA-seq datasets from different laboratories, the integrated data may result in a gene having multiple names. To eliminate this issue, we used the gene list compiled by the HUGO Gene Nomenclature Committee (HGNC) to merge the expression matrices of the same gene, ensuring that each gene is mapped to a unique HGNC symbol. Potential T cells were first isolated based on the positive expression of CD3 (including CD3D, CD3G, and CD3E). The data were then normalized using *scanpy.pp.normalize_total* function, log transformed with *scanpy.pp.log1p* function and highly variable genes (HVGs) identified using the *scanpy.pp.highly_variable_genes* function from Scanpy package (version 1.9.8). Next, principal component analysis (PCA), Batch balanced KNN (BBKNN), uniform manifold approximation and projection (UMAP), and Leiden clustering were applied for downstream analyses using the Scanpy algorithm. After unsupervised clustering, we manually excluded clusters expressing marker genes of other cell types, such as B cells, myeloid cells and epithelial cells. Based on the expression of CD4 and CD8 genes, T cell clusters were annotated as $CD4^+$, $CD8^+$, and $CD4^-CD8^-$ double negative T (DNT) subsets.

Notably, this study focused exclusively on $CD3^+CD4^-CD8^-$ DNT cells, while other DNT cell subtypes were not considered. Furthermore, to avoid false positives, we retained only those cells within the DNT subset that showed negative expression of CD4 and CD8 genes (*CD4*, *CD8A* and *CD8B*). We performed another round of unsupervised clustering, following the same steps as described above. Markers were determined by *scanpy.tl.rank_genes_groups* function (using the Wilcoxon test) of Scanpy packgae.

Metrics to evaluate scRNA-seq data integration quality

To evaluate the quality of pan-cancer scRNA-seq data integration, we employed multiple assessment framework: (1) the Local Inverse Simpson's Index (LISI) algorithm based on the LISI R package [47]; and (2) a suite of evaluation metrics from the scIntegrationMetrics R package, including iLISI and the Average Silhouette Width by celltype (celltype ASW) [48, 49]. The "integration LISI" (iLISI) measures the effective number of batches in a local neighborhood, with higher values indicating better batch mixing. Celltype ASW is defined as a measure that compares the intra-cluster distances of cells within the same cell type to their inter-cluster distances from cells of other types; higher ASW values reflect better preservation of biological structure. The ideal integration maximizes batch mixing while fully retaining the inherent biological architecture.

DNT subsets annotation

To investigate the transcriptional programs of each DNT cell subtype, we compared each DNT subset with $CD4^+$ T cells, $CD8^+$ T cells, and other DNT cell clusters, and identified differentially expressed genes (DEGs) using *scanpy.tl.rank_genes_groups* function (using the Wilcoxon test) from Scanpy package. We selected intersection of DEGs in each DNT subset with $CD4^+$ and $CD8^+$ T cells for pathway enrichment analyses (table S3).

Shannon equitablity index calculating

Following Liangtao Zheng et al. [27], to compare the distribution of DNT cell subtypes among blood, normal, adjacent and tumor samples, we independently quantified the proportions of αβ DNT and γδ T cell subsets. Then the diversity of these distributions was assessed using the Shannon equitability index (EH), a normalized variant of the Shannon diversity index, defined by the equation:

$$Shannon\ Equitablity = -\frac{\sum_{i=1}^{N} p_i log_2 p_i}{log_2(N)}$$

In this formula, p_i represents the relative frequency of each DNT subset, and N denotes the total number of DNT subsets evaluated. The EH metric ranges from 0 to 1, where values approaching 1 indicate a balanced distribution of subsets, and lower values signify that

certain subsets are predominantly represented within the sample.

Distribution of DNT subsets across tissues and cancer types

To evaluate the distribution patterns of DNT subsets across different cancer types, we determined the proportion of each DNT subset relative to the overall DNT population in samples from various cancer types. Any cancer type that contained fewer than 200 cells was excluded to enhance the reliability of our estimates and mitigate potential biases. To assess the enrichment of individual DNT subsets in distinct pathological states, the ratio of observed to expected cell counts (*Ro/e*) for each subset was calculated across multiple tissue types. Samples were categorized into four groups according to tissue origin: Normal, Adjacent, Blood, Tumor. "Normal" refers exclusively to solid, non-blood tissues and comprises two sources: (1) solid tissues from healthy donors, and (2) distant, histologically non-involved tumor-adjacent tissues as annotated in the originating studies. Expected cell counts were estimated using chi-square tests to establish the baseline distribution of DNT subsets, with enrichment defined as $Ro/e > 1$.

Reconstruction the TCR repertoires of DNT cells

TCR CDR3 sequences from both αβ DNT and γδ T cells, were reconstructed using TRUST4 v.1.1.3 algorithm (https://github.com/liulab-dfci/TRUST4). BAM files were used as input for TRUST4, which realigns the contigs to IMGT reference sequences to identify the corresponding gene and CDR3 details.

TCR diversity and evenness index

TCR diversity was calculated by Shannon entropy using the R package vegan (version 2.6.8) (https://cran.r-project.org/web/packages/vegan/index.html). The formulation of Shannon entropy to the:

$$shannon\ entropy = -\sum_{i=1}^{N} p_i log p_i$$

where pi was the frequency of CDR3 sequence in the TCR repertoire and N was the total number of unique clonotypes. A larger value indicates a greater diversity.

Gini-coefficient represents the evenness of TCR distribution. We calculated Gini-coefficient using Gini function in R package DescTools (version 0.99.47) (https://cran.r-project.org/web/packages/DescTools/index.html). Gini Coefficient ranges between 0 (all clones are equally frequent) and 1 (frequency dominated by one clone).

Gene set enrichment analysis

The enrichGO function from the clusterProfiler R package (version 4.10.1) was applied to perform enrichment analysis on all differential genes obtained, where pathways with a p value < 0.05 were recognized to be significantly enriched.

Calculation of gene signature scores

All gene set scores were computed based on the *scanpy.tl.score_genes* function in scanpy, which represented the value obtained by subtracting the average expression of genes in input gene set from the average expression of genes in the reference gene set selected by random sampling. The gene signatures used in this study were shown in table S5.

Transcription factors activity analysis

The python version of SCENIC (single-cell regulatory network inference and clustering) was used to compute gene regulatory networks for DNT single-cell transcriptome data. Transcription factors associated with each cellular state of DNT were identified based on co-expression and motif analysis. Briefly, it was performed in three steps: inferring co-expression modules, identifying regulons, and calculating regulon activity. Finally, the regulon specificity score (RSS) for each DNT subset were calculated and ranked. All parameters were set to default values.

Cellular response to signaling molecules of DNT

To explore the characteristics of cytokines secreted by each subtype of DNT, we predicted the intracellular signalling responses to 43 cytokines for each subtype based on the CytoSig platform. Expression matrix of single cells was input in Cell Ranger output format, the number of randomizations in the permutation test was set to default value 1000, the minimal read counts required for each barcode was set to 0, maximal zero dropout rates allowed for each gene was set to 1. Based on the output of scaled activity score, we calculated the average activity score of each cytokine for each DNT subset.

Antigen presentation function analysis of T14_HLA-DR+, DCs and macrophages

To compare the differences in antigen presentation function between T14_HLA-DR$^+$ and DC cells, we characterized the antigen presentation process of both based on SOM analysis. The gene expression similarities of T14_HLA-DR$^+$ and DCs were projected into a two-dimensional heatmap, where each unit (metagene) represents clusters of co-expressed genes across single cells. The size of the square SOM grid was set to 60×60 metagenes, read count preprocessing were set to be activated, other parameters were set to default values. The SOM analysis

was based on Single Cell R-Analysis Tools (SCRAT) (version 2.1.0), whose pipeline included data preprocessing, SOM training, metagene heatmap visualization, and gene set enrichment analysis.

DNT lineage trajectory inference and RNA velocity

To reconstruct the αβ T and γδ T cell lineage trajectories along the DNT cell differentiation process, several state-of-the-art cell trajectory inference algorithms were employed, including scVelo, CellRank, Dynamo, scTour. For RNA velocity, .bam files from Cell Ranger output were used to extract spliced and unspliced RNA data, thus generating .loom files. After converting .loom files to .h5ad format, transcriptional, splicing, and degradation rates were estimated using the scVelo, CellRank and Dynamo pipeline. Specifically, scVelo infers transient cell states by fitting dynamic ordinary differential equations; CellRank uses Markov chain transition probabilities to estimate fate potentials; and Dynamo reconstructs continuous vector fields that can additionally incorporate metabolic labeling information to estimate absolute rates. For cells without raw data, scTour were used to infer differentiation trajectories, and pseudo times were calculated to aid validation. Cells were then ordered along pseudotime and visualized in UMAP space.

Diffusion pseudotime analysis

To characterize DNT subset transitions, the diffusion map algorithm was employed to preserve global cell relationships and infer pseudotemporal ordering of cells, thereby elucidating differentiation trajectories. Utilizing the batch-aligned neighborhood graph generated by BBKNN, we constructed the diffusion map with the *scanpy.tl.diffmap* function. Root cells were identified using Dynamo on the diffusion map, and diffusion pseudotime was subsequently calculated with the *scanpy.tl.dpt* function. Finally, we applied a generalized additive model to correlate individual gene expression profiles with pseudotime.

Cell type deconvolution of bulk RNA-seq and spatial transcriptome data

For bulk RNA sequencing data, the docker version of CIBERSORTx was utilized to estimate cell type fractions within TCGA (The Cancer Genome Atlas) datasets and immunotherapy cohorts. Briefly, for each bulk RNA-seq cohort, we first applied CIBERSORTx with the built-in LM22 signature matrix to estimate total T-cell abundance. Next, our single-cell data were used as a reference to infer the abundance of DNT subtypes. A random sample of 100,000 cells was taken for all T cells and a reference set was created using their count matrix. The S-mode batch correction mode was adopted with other parameters set to default values. Finally, imputed T-cell fractions were normalized by each sample's total T-cell abundance. These deconvolution results were used for downstream DNT immune typing and survival analyses.

For spatial transcriptomic data, the relative abundance of different cell types was estimated by the conditional autoregressive deconvolution (CARD) method. Spatial transcriptomics count matrix paired with spatial location coordinates were inputted to infer cell type composition. The cell type and sample information for each cell was used to create a reference for deconvolution. Ligand-receptor (LR) pair expression intensity was assessed based on the average expression of corresponding genes. The LR pair gene set scores were calculated to evaluate intercellular communication within spatial regions. For Visium HD datasets, we employed spot deconvolution to classify and annotate spatial bins with cell types identified from the single-cell atlas. Utilizing the spacexr package (version 2.2.1) in doublet mode, we constrained each spatial bin to contain no more than two distinct cell types.

Cell-cell interaction analysis

To elucidate the differences in cell-cell interactions between different DNT cells and other immune cell types within our dataset, the LIANA + framework was employed to systematically infer ligand-receptor interactions. It functions as an integrative framework that harmonizes multiple widely used methods, along with their associated ligand-receptor interaction databases. In this study, we used the CellChat method implemented within LIANA in combination with Omnipath's extensive ligand-receptor database to capture cell-cell communication event separately in tumor and adjacent tissues. The strength and type of interactions in different tissues were compared and summarized. LR pairs with statistically significant interactions ($p < 0.05$) were identified and extracted for further analysis. Minimum expression proportion for the ligands and receptors was set to 0.1.

Hypoxia activity evaluation

The activity of the hypoxia pathway was evaluated on three distinct levels: (1) Hypoxia pathway activity predicted by PROGENy (version 1.17.3). PROGENy infers pathway activities from gene expression data by leveraging a collection of validated gene signatures, providing a robust estimation of pathway activation levels; (2) Quantified score of hypoxia-related metabolic pathways by scMetabolism package (version 0.2.1), the quantification method was set to VISION; (3) HIF-1α activity assessed by DoRothEA gene regulatory network which comprises signed transcription factor-target gene interactions. The decoupleR package (version 2.8.0) was used to infer HIF-1α activity based on this network.

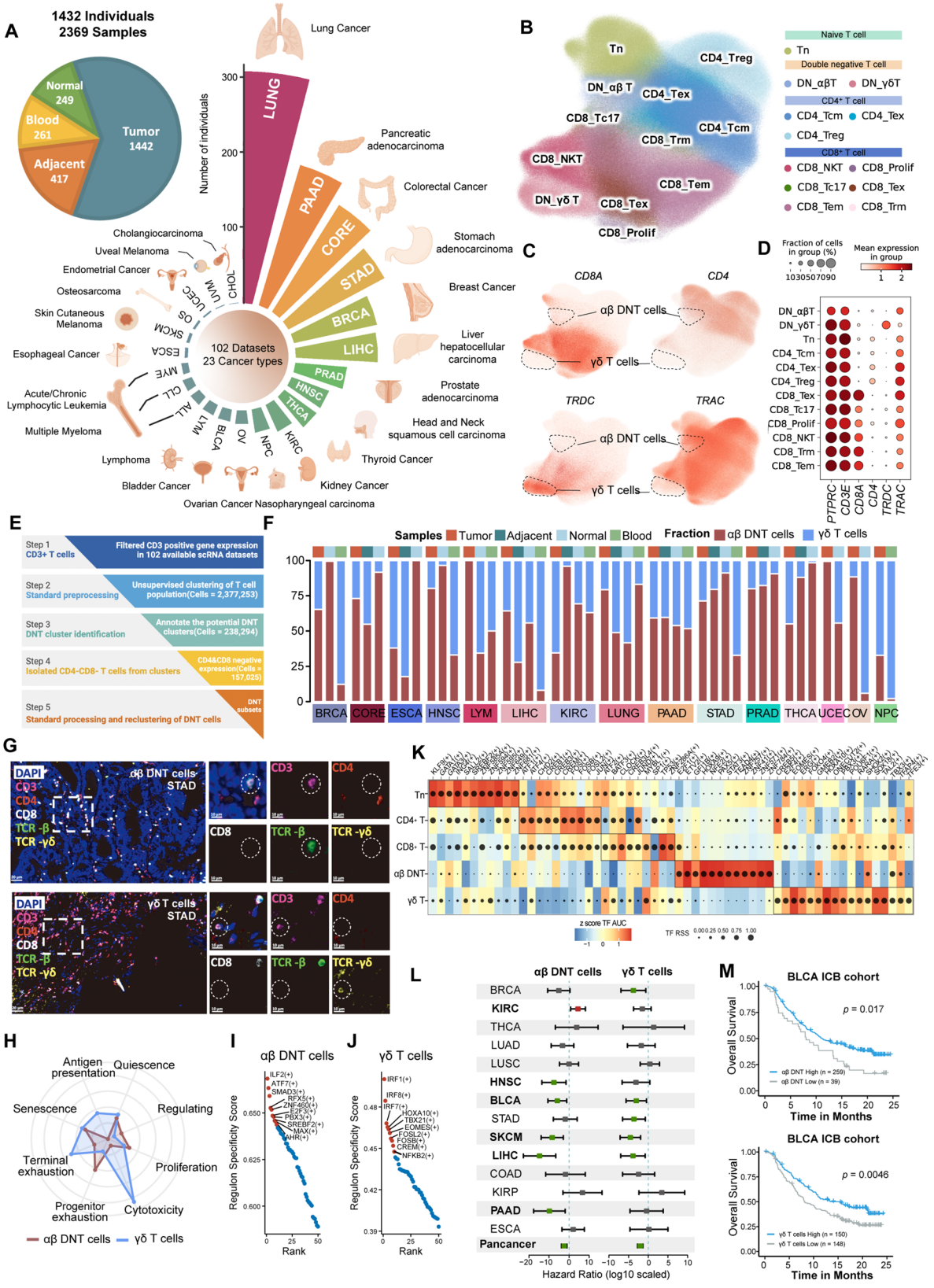

Fig. 1 (See legend on next page.)

(See figure on previous page.)

Fig. 1 Single-cell transcriptomic atlas of DNT cells. **A** Graphs depicting the various cancer types and sample composition involves in this study. **B** and **C** UMAP plot of the 2,377,253 T cells, colored by (**B**) associated cell types or (**C**) specific marker genes. **D** Dot plot showing selected gene expression. Dot size indicates the percentage of cells expressing the listed genes. Red color indicates mean expression level within group. **E** Flow diagram illustrating the process used to obtain the high-quality DNT cells. **F** Stacked bar plots showing the proportions of two DNT cell types across cancer and sample types. Only tumor types with more than 500 DNT cells and containing more than one sample type were included. The top annotation bar indicates tissue types: red represents tumor tissues, dark blue represents adjacent tissues, light blue represents normal tissues, and green represents blood. The bottom annotation bar represents cancer types, with each color corresponding to a specific tumor type. **G** Representative example of STAD samples stained by multiplex immunofluorescence to show αβ DNT (top) and γδ T (bottom) cells (dashed circles). Scale bars, 10 μm. **H** Radar plot showing the functions of αβ DNT (red) and γδ T cells (blue). **I** and **J** The αβ DNT (I) and γδ T cell specific regulons (**J**) ranked by regulon specificity scores (RSS) in descending order. The top ten regulons are colored by red. **K** Heatmap showing transcription factor expression of regulons. Dot size represents cell-type specificity RSS. **L** Forest plot of hazard ratio calculated by multiple Cox proportional hazards models with overall survival, adjusted for tumor stage, age and sex in the TCGA cohorts. Hazard ratios with 95% confidence interval are depicted. The p values calculated by Wald test. Red block indicates the worse survival ($p < 0.05$); green block represents better survival ($p < 0.05$); grey block indicates $p > 0.05$. **M** Kaplan-Meier overall survival curves for BLCA ICB cohorts grouped by αβ DNT (top) and γδ T cells (bottom). *p* values calculated by log-rank test. Optimal cut-offs (αβ DNT: 2.07%; γδ T: 8.58%) were determined by the surv cutpoint function (> cut-off = High, ≤ cut-off = Low). The DNT proportions were adjusted for total T cell abundance. * *p* value < 0.05, ** *p* value < 0.01, *** *p* value < 0.001, and **** *p* value < 0.0001

Identification and random forest classification for DNT immune typing

To identify potential different patterns of DNT cell compositions in distinct TMEs across various cancer types, we quantified the proportion of each DNT subset within individual tumor samples. Based on the frequency of these subsets, 2 main immune types ($\alpha\beta T^{Const}\gamma\delta T^{Flux}$ and $\alpha\beta T^{Flux}\gamma\delta T^{Const}$) were identified by hierarchical clustering with Euclidean distance as distance metric. Subsets with low frequency in most samples were excluded from the clustering. The ComplexHeatmap (version 2.18.0) R package was used for visualization.

To investigate the prognostic significance of the two DNT immune types, we trained a random forest (RF) classifier with the randomForest R package (version 4.7.1.1) using 385 patients from our DNT dataset, where per-individual features comprised the relative frequencies of DNT subsets, along with age, sex, tumor stage and tumor type as nuisance covariates. The dataset was trained with a 70/30 train/held-out split, hyperparameters tuned on the training set via out-of-bag error (final ntree = 1000, mtry = 9, importance = TRUE, proximity = TRUE), with fixed random seeds; class weights were applied where appropriate. To avoid information leakage from tumor-type-specific structure, model evaluation used blocked/stratified cross-validation by tumor type/cohort, and variable importance reflects permutation-based estimates under this scheme.

Before applying the trained RF to TCGA, we filtered TCGA samples to ensure compatibility and complete covariates: (i) we excluded tumor types not represented in our single-cell reference (to avoid out-of-domain predictions), and (ii) we removed samples with incomplete clinical data (e.g., missing age/sex, stage). After screening, 7,033 TCGA tumor samples were retained for analysis. For each patient, the adjusted DNT frequencies obtained from CIBERSORTx deconvolution with the built-in LM22 signature matrix. The age, sex, tumor stage and tumor type, were used as input features to predict the corresponding class ($\alpha\beta T^{Const}\gamma\delta T^{Flux}$ and $\alpha\beta T^{Flux}\gamma\delta T^{Const}$). The predicted classes were subsequently utilized to assess overall survival differences.

Survival

To evaluate the impact of various DNT cell types on overall survival, we applied Cox proportional hazards models and Kaplan-Meier analysis across the TCGA and immunotherapies cohorts. In the Cox models, the T-cell-adjusted DNT frequency derived by CIBERSORTx (details see 'Cell type deconvolution of bulk RNA-seq and spatial transcriptome data') served as the primary exposure variable, with age, sex, and tumor stage included as covariates. Robust (sandwich) standard errors were used, proportional hazards assumptions were verified, and multiple testing across tumor types/subsets was controlled by BH-FDR. Forest plots were created with the forestplot (version 3.1.3) R package.

For Kaplan-Meier analysis, optimal cut-off points for each DNT subtypes were identified using the *surv_cutpoint* function from the survminer (version 0.4.9) R package. Based on these cut-offs, patients were stratified into high- and low-infiltration groups. To visualize survival differences, Kaplan-Meier overall curves were conducted using survival (version 3.7.0) and survminer (version 0.4.9) R packages.

Results

Single-cell profiling of double negative T (DNT) cells

Due to the limited number of DNT cells in each sample, we compiled 102 single-cell RNA datasets, comprising both publicly available and in-house data to construct a comprehensive pan-cancer single-cell transcriptomic atlas of DNT cells (table S1). These datasets include 2,369 samples from 1,432 individuals across 23 cancer types, covering tumor tissues, adjacent non-tumor tissues, healthy controls and peripheral blood from cancer patients (Fig. 1A). To avoid the study bias, at least two datasets were included for each cancer type. Following

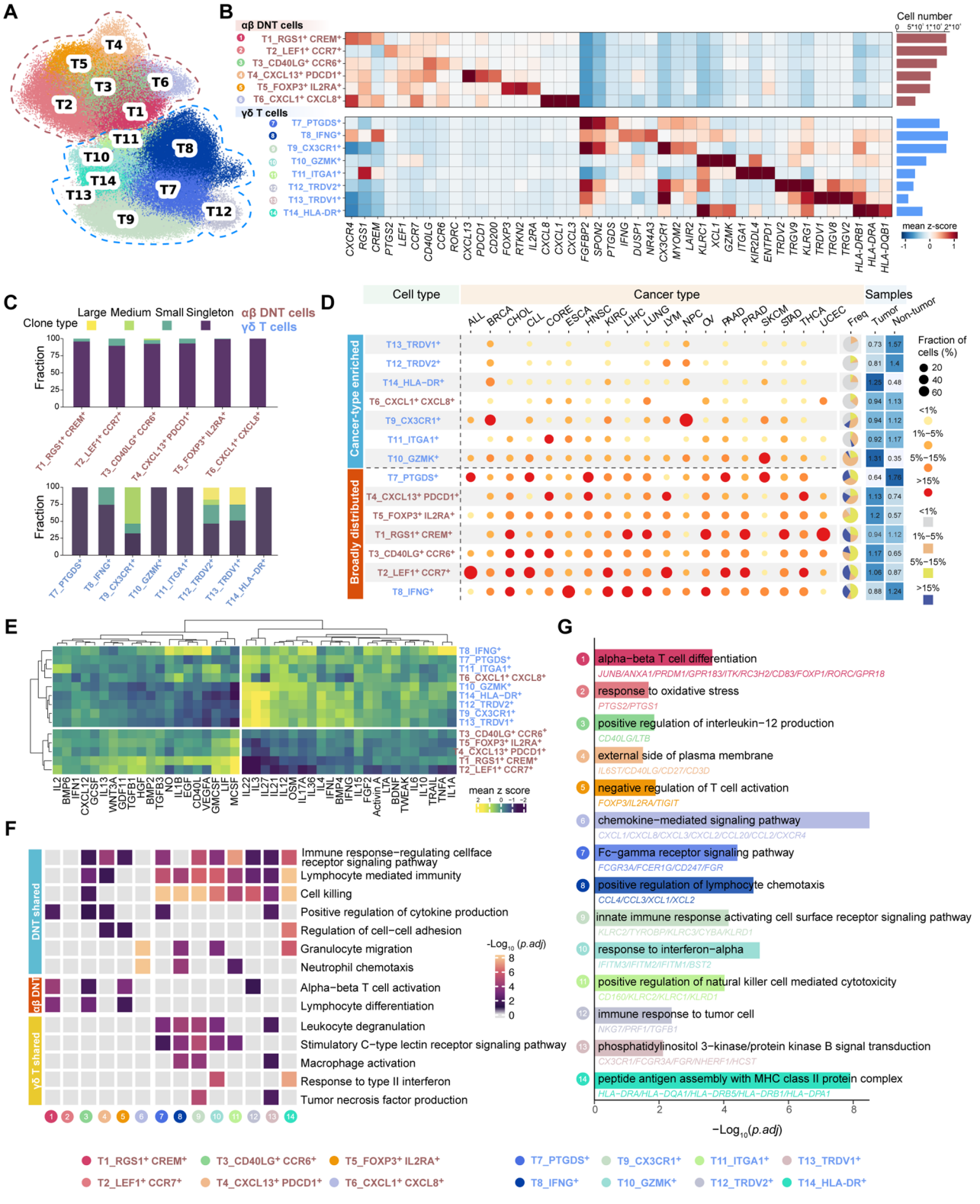

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Fig. 2 Transcriptional signatures, heterogeneity and functional diversity of DNT subsets. **A** The UMAP plot of 157,025 single DNT cells, colored by associated cell types. **B** Heatmap displaying the highly expressed genes for each DNT subsets. The bar plot indicates their frequency after square-root transformation. **C** Bar plot illustrating the clonotype distribution of each cell type, categorized as Singleton ($n = 1$), Small ($1 < n < 5$), Medium ($5 < n < 10$), and Large ($10 < n < 100$). **D** Dot plot depicting the frequency of 14 DNT subsets across multiple cancer types (left). Pie plots indicate the abundance distribution of each cell type across different tumor types (middle). Heatmap representing the Ro/e ratio in tumor and non-tumor samples (right). Red labels indicate αβ DNT subgroups, while blue labels indicate γδ DNT subgroups. Cancer types contributing < 200 total DNT cells to the integrated atlas were excluded. **E** Heatmap depicting the cytokine activities of DNT subsets predicted by CytoSig. **F** Heatmap showing the shared GO enrichment terms for DNT subsets. The color indicates adjusted *p* value. **G** GO enrichment of biological processes for unique DNT subsets. Bar lengths indicate statistical significance of the enrichment (adjusted *p* value)

rigorous quality control, we identified a total of 2,377,253 high-quality $CD3^+$ T cells, which were subjected to batch effect correction and unsupervised clustering. This analysis revealed 12 distinct subclusters based on canonical markers, including Naïve T (Tn), $CD8^+$ T cells (CD8_NKT, CD8_Trm, CD8_Tem, CD8_Tex, CD8_Tc17, and CD8_prolif), $CD4^+$ T cells (CD4_Tcm, CD4_Tex, CD4_Treg), and $CD4^-CD8^-$ DNT cells (αβ DNT and γδ T cells) (Fig. 1, B to D, and fig. S1, A to C). Focusing specifically on DNT cells, we further excluded any cells expressing CD4 or CD8 from these two clusters, yielding 157,025 high-quality single DNT cells for subsequent profiling analysis (Fig. 1E).

Significant heterogeneity was observed in the distribution of αβ DNT and γδ T cells across different cancers and tissue types. For instance, the ratio of αβ DNT to γδ T cells in tumor samples was low in kidney cancer (KIRC) and esophageal cancer (ESCA) but high in uterine corpus endometrial carcinoma (UCEC), regardless of the dataset derived (Fig. 1F and table S2). In lung cancer (LUNG), the proportion of γδ T cells decreased progressively in the order of normal, adjacent, and tumor, whereas the opposite trend was observed in gastric cancer (STAD) and prostate cancer (PRAD) (Fig. 1F and fig. S1D). The distribution patterns of these two DNT cell subsets were not reflected in peripheral blood samples from cancer patients, suggesting that blood may not accurately represent tumor-infiltrating DNT cells (Fig. 1F). Through multiplex immunofluorescence (mIF), the presence of infiltrating γδ T cells and αβ DNT cells was validated in several cancer types, including STAD, colorectal cancer (CORE) and LUNG (Fig. 1G and fig. S1E).

Functionally, γδ T cells exhibited strong cytotoxicity and partial antigen-presenting capabilities, while αβ DNT cells were predominantly involved in regulatory functions, as annotated by the TCellSI [50] (Fig. 1H). This functional diversity was further supported by their transcription factor (TF) profiles. For instance, regulons enriched in αβ DNT cells were associated with cell cycle regulation/proliferation (e.g., *E2F3* [51]), and immune modulation (e.g., *AHR* [52]) (Fig. 1I). In contrast, γδ T cells displayed elevated regulons linked to immune activation (e.g., *IRF1*, *IRF7*, *IRF8*, *EOMES*, and *TBX21*) and inflammation (e.g., *NFKB2*) (Fig. 1J). Collectively, each T cell subcluster exhibited distinct regulon profiles (Fig. 1K). These results highlight the functional specialization of αβ DNT and γδ T cells in the tumor microenvironment (TME) and their heterogeneity across cancer types and tissue origins.

Furthermore, deconvolution of bulk transcriptome data revealed that an elevated proportion of both αβ DNT cells and γδ T cells correlates with improved prognosis across multiple TCGA cancer types and immunotherapy cohorts (Fig. 1, L and M, and fig. S2, A to C). To validate these findings, we established subcutaneous tumor models using MC38 murine colon cancer cells. Mice were randomly divided into four experimental groups: (1) γδ T cell-depleted (aγδT) and anti-PD-1 treatment; (2) γδ T cell-depleted (aγδT) and an isotype IgG; (3) PBS control and anti-PD-1 treatment; (4) PBS control and an isotype IgG (fig. S2D). Anti-PD-1 treatment in the PBS control group significantly suppressed tumor growth (fig. S2E). However, specific depletion of γδ T cells significantly compromised the therapeutic efficacy of anti-PD-1 treatment (fig. S2E). These results suggested that γδ T cells play a pivotal role in immunotherapy.

Transcriptional signatures, heterogeneity and functional diversity of DNT subsets

Unsupervised clustering of DNT cells identified 6 αβ DNT (T1 to T6) and 8 γδ T cell subsets (T7 to T14) (Fig. 2A and fig. S3). These subsets exhibited specific/enriched signature genes, which were used to define their identities (Fig. 2B and table S3). Consistent with previous studies on DNT cells focusing on non-tumor contexts using experimental approaches such as flow cytometry, our analysis confirmed the presence of several reported subsets across different cancer types. For αβ DNT cells, these included the regulatory subset (T5_$FOXP3^+IL2RA^+$) [12–15], the high PD-1 expression subset (T4_$CXCL13^+PDCD1^+$) [53], the RORC-expressing subset (T3_$CD40LG^+CCR6^+$) [54], the naïve subset (T2_$LEF1^+CCR7^+$) [55], and the $CD8^+$ T cells-derived subset (T1_$RGS1^+CREM^+$) [56]. For γδ T cells, the identified subsets included the IFNγ-producing subset (T8_$IFNG^+$) [16, 17], the granzymes K positive subset (T10_$GZMK^+$) [18], the major histocompatibility complex (MHC)-II-expressing subset (T14_HLA-DR^+) [11, 19], gut resident subset (T11_$ITGA1^+$) [25], fractalkine receptor positive subset (T9_$CX3CR1^+$) [57], and the traditionally

classified TCR-based subtypes: Vδ1$^+$ (T13_TRDV1$^+$) [58, 59] and Vδ2$^+$ (T12_TRDV2$^+$) [60, 61] (Fig. 2B). In addition, two previously uncharacterized subsets were identified, including T6_CXCL1$^+$CXCL8$^+$ of αβ DNT and T7_PTGDS$^+$ of γδ T cells (Fig. 2B). T6_CXCL1$^+$CXCL8$^+$ was characterized by elevated expression of chemokines, suggesting a potential role in chemotaxis, while T7_PTGDS$^+$ was distinguished by high levels of PTGDS and FGFBP2, indicating effector functions. The comprehensive transcriptomic profiling of these subsets enabled us to investigate the shared and distinct role of DNT cell-mediated immune responses across various cancer types.

Next, we reconstructed immune receptor repertoires for both αβ DNT and γδ T cells using TRUST4 [62]. Most αβ DNT subsets were singleton with similar Gini coefficients (Fig. 2C and fig. S4A). Among them, T2_LEF1$^+$CCR7$^+$ exhibited the highest Shannon diversity score, reflecting its naïve features (fig. S4A). In contrast, γδ T subsets, particularly T12_TRDV2$^+$ and T13_TRDV1$^+$, showed significant TCR clonal expansion (Fig. 2C). Regarding VDJ recombination, T12_TRDV2$^+$ was predominantly composed of Vδ2 and Vγ9 chains, while T13_TRDV1$^+$ primarily composed Vδ1, aligning with their respective gene expression profiles (fig. S4B). Consistent with previous research on TCR recombination and selection, αβ DNT subsets showed no significant biases toward VDJ usage, except for TRBV20-1 and TRBJ2-7 (fig. S4C). We further explored the distribution of DNT cell subsets across different cancer types. Notably, while most αβ DNT subsets demonstrated broadly distribution (excluding T6_CXCL1$^+$CXCL8$^+$), γδ T cell subsets showed predominant cancer-type enrichment, except for T7_PTGDS$^+$ and T8_IFNG$^+$ (Fig. 2D). Particularly, T10_GZMK$^+$ was enriched in SKCM, T11_ITGA1$^+$ in CORE, and T9_CX3CR1$^+$ in BRCA and NPC (Fig. 2D). Tissue preferences were also observed, with tumor samples showing greater diversity in both infiltrating αβ DNT and γδ T cell compositions (measured by Shannon equitability index, fig. S4D). In additional, specific subsets enriched in tumor samples (e.g., T14_HLA-DR$^+$, and T10_GZMK$^+$) or non-tumor samples (e.g., T7_PTGDS$^+$ and T13_TRDV1$^+$) (Fig. 2D).

To delineate the functional heterogeneity of DNT cells, we first assessed cytokine signaling activities using CytoSig [63], uncovering distinct cytokine profiles between αβ DNT and γδ T cells. αβ DNT cells were predominantly associated with immune cell activation (e.g., LIF) and regulation (e.g., MCSF, GMCSF), whereas γδ T cells exhibited a dual functional role, inducing both anti-tumor (e.g., IL-27, IL-12, IL-21, and IFNγ) and pro-tumor (e.g., IL-22, IL-17 A, IL-4) cytokine activities [64] (Fig. 2E). Particularly, T8_IFNG$^+$ subset demonstrated remarkable specificity for TNFA and TRAIL, indicating direct anti-tumor effects. These distinct cytokine profiles underscored the diverse and dynamic roles of αβ DNT and γδ T cells in tumor immunity. Next, given that different T cell states were shared across DNT subsets (fig. S5), Gene Ontology (GO) enrichment analysis revealed shared functions among DNT subsets (e.g., lymphocyte mediated immunity), as well as αβ DNT- (e.g., α-β T cell activation) and γδ T- specific function (e.g., stimulatory CLR signaling pathway) (Fig. 2F). Within the shared function, all γδ T subsets but only T3_CD40LG$^+$CCR6$^+$ among αβ DNT subsets exhibited cell killing signaling, which is consistent with enriched cytotoxicity state in γδ T subsets (Fig. 2F and fig. S5). On the other hand, distinct functional roles emerged within individual αβ DNT and γδ T cell subclusters. For αβ DNT cells, only T1_RGS1$^+$CREM$^+$ represented a terminal differentiation state, while the newly characterized T6_CXCL1$^+$CXCL8$^+$ subset predominantly facilitated chemokine-mediated leukocyte recruitment, with the highest statistical significance observed (Fig. 2G). For γδ T cells, which span both innate and adaptive immunity [65], displayed diverse functional specialization. For instance, the T14_HLA-DR$^+$ demonstrated strong antigen-presenting capabilities, consistent with its T cell states (Fig. 2G and fig. S5), while the novel subset T7_PTGDS$^+$ may induce tumor cell death via antibody-dependent cell-mediated cytotoxicity through Fc-gamma receptors [64] (Fig. 2G). These findings reveal a nuanced functional diversity within DNT cells, emphasizing their critical roles in tumor immunity and highlighting the complexity of DNT-mediated immune responses.

Evolutionary dynamics of tumor DNT cells

Next, we employed multiple computational tools (i.e., Dynamo [66], scvelo [67] and CellRank [68]) to uncover potential lineage relationships among DNT subsets within tumor samples. Evolutionary dynamics revealed distinct differentiation trajectories originating from the T2_LEF1$^+$CCR7$^+$ subset for αβ DNT cells and the T10_GZMK$^+$ subset for γδ T cells, respectively (Fig. 3A, and fig. S6, A and B). To better preserve the differentiation trajectories, we applied diffusion map to construct coordinates along with velocity [69], which revealed a differentiation trajectory of αβ DNT cells from T2_LEF1$^+$CCR7$^+$ towards either T1_RGS1$^+$CREM$^+$ or T4_CXCL13$^+$PDCD1$^+$ (Fig. 3B). For γδ T cells, two potential trajectories were identified: Trajectory 1 (Traj1) featured velocity vectors originating from T10_GZMK$^+$ to T9_CX3CR1$^+$ via intermediate states T14_HLA-DR$^+$ and T13_TRDV1$^+$; Trajectory 2 (Traj2) extended from T10_GZMK$^+$ to T12_TRDV2$^+$ via T7_PTGDS$^+$ and T8_IFNG$^+$ (Fig. 3C). These trajectories were further supported by the overlap of TCR sequences (Fig. 3D). Moreover, Traj1 was characterized by a pronounced upregulation of *TRDV1* expression along pseudotime with stable *TRDV2*

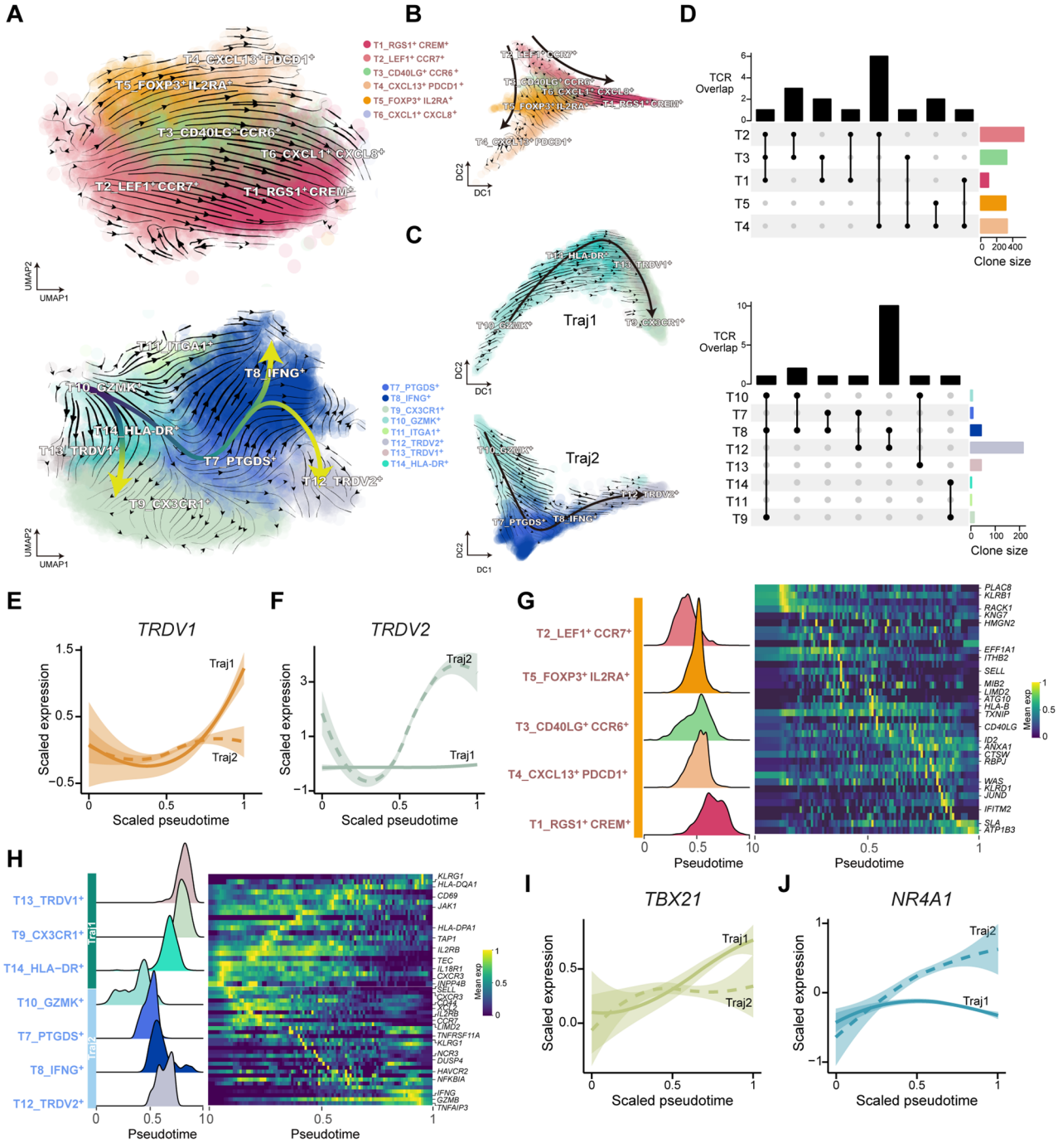


Fig. 3 Distinct differentiation linages of DNT cells. **A** RNA velocity streamline plots using Dynamo showing the differentiation linages of αβ DNT (top) and γδ T cell (bottom) subtypes on UMAP projection, colored by cell types. **B** and **C** RNA velocity streamline plots visualizing the trajectory of αβ DNT cell (B) and γδ T cell (**C**), including Traj1 (top) and Traj2 (bottom), on the diffusion map projection. DC, Diffusion component. **D** Upset plot illustrating the shared TCR clones among multiple αβ DNT cell (top) and γδ T cell (bottom) subtypes. Black bars indicate the number of overlapping TCRs detected. **E** and **F** Expression of TRDV1 (**E**) and TRDV2 (**F**) along the pseudotime. The solid and dashed lines represent Traj1 and Traj2, respectively, with the shaded area indicating the 95% confidence interval. **G** (Left) Density plot showing the distribution of αβ DNT cell subsets along the pseudotime. (Right) Heatmap showing differentially expressed genes over pseudotime. The pseudotime is scaled from 0 (the earliest) to 1 (the latest). **H** (Left) Density plot showing the distribution of each γδ T cell subsets along the pseudotime. (Right) Heatmap depicting differentially expressed gene over pseudotime. Both trajectories start from T10_GZMK$^+$ and extend towards T13_TRDV1$^+$ (Traj1) and T12_TRDV2$^+$ (Traj2), respectively. The pseudotime is scaled from 0 (the earliest) to 1 (the latest). **I** and **J** Expression of TBX21 (**I**) and NR4A1 (**J**) along the pseudotime. The solid and dashed lines represent Traj1 and Traj2, respectively, with the shaded area indicating the 95% confidence interval

expression (Fig. 3E), while Traj2 exhibited the inverse pattern (Fig. 3F).

To investigate the underlying mechanisms, we first analyzed dynamic gene expression changes along the pseudotime. In αβ DNT cells, early stages were marked by the expression of immune activation genes (e.g., *KLRB1*), whereas late stages displayed elevated expression of differentiation regulators (e.g., *ID2*) and immune homeostasis genes (e.g., *SLA*) (Fig. 3G). For γδ T cells, early-stage genes were primarily linked to activation (e.g., *SELL*) and proliferative responses (e.g., *CD44*). In Traj1, mid-to-late stages showed enrichment for genes involved in antigen presentation (e.g., *TAP1*, *HLA-DPA1*, and *HLA-DQA1*) and effector functions (e.g., *KLRG1*), while Traj2 exhibited upregulation of cytotoxicity-related genes (e.g., *IFNG*, *GZMB*), exhaustion markers (e.g., *HAVCR2*) and anti-inflammatory responses (e.g., *TNFAIP3*) (Fig. 3H). Next, we compared transcription factor profiles between the two trajectories (fig. S6C). Traj1 was enriched for factors associated with enhanced effector functions and tumor cytotoxicity (e.g., TBX21) (Fig. 3I). In contrast, NR4A1, a transcription factor implicated in negative regulation of T cell activation and exhaustion [70, 71], was significantly elevated in Traj2 (Fig. 3J and fig. S6C). Therefore, Traj1 progresses toward a cytotoxic phenotype, while Traj2 appears to follow a trajectory toward an exhausted or immunosuppressive state. Collectively, these findings highlight the distinct differentiation trajectories and regulatory mechanisms governing αβ DNT and γδ T cell fates in the TME.

Interaction profiles of DNT cells and antigen-presenting characteristics of γδ T cells

To further elucidate the function of DNT cells, we conducted cell-cell communication analysis to examine their interactions within the TME. Notably, γδ T subsets exhibited stronger interaction intensity with $CD4^+$/$CD8^+$ T cells and various myeloid cell subtypes compared to αβ DNT subsets (Fig. 4A and fig. S7A). Specifically, the T14_HLA-DR^+ subset demonstrated the most prolific interactions, particularly with $CD8^+$ Trm, $CD8^+$ Tex, $CD8^+$ Tem, and $CD4^+$ Tex cells (Fig. 4A). Previous studies have shown that γδ T cells can cross-present both MHC class I- and MHC class II-restricted peptides, effectively facilitating the activation and proliferation of conventional αβ T cells [11, 72, 73]. Given its high expression of MHC class II genes, the T14_HLA-DR^+ subset was thus identified as an antigen-presenting cell (APC) involved in MHC class II-restricted peptide processing and presentation (Fig. 4B and fig. S7B). To gain deeper insights into the functional properties of T14_HLA-DR^+ cells and other classical APCs (e.g., dendritic cells (DCs) and macrophages), we performed a self-organizing map (SOM) analysis [74] (Fig. 4C). While DCs and macrophages exhibited similar key functions such as antigen processing (e.g., protein ubiquitination in DCs, digestion in macrophages), antigen trafficking (e.g., vesicle-mediated transport in DCs, extracellular vesicle biogenesis in macrophages), and antigen presentation, the T14_HLA-DR^+ subset displayed unique mechanisms (Fig. 4C). Specifically, T14_HLA-DR^+ cells bound antigenic peptides via MHC II molecules and utilized endocytic cycling to efficiently display antigens, thereby activating T cells (Fig. 4C). Beyond antigen-presenting, T14_HLA-DR^+ can modulate T cell activation by influencing the intensity of T cell receptor signaling and also demonstrated direct cytotoxicity and cell killing activity (Fig. 4, B and C, and fig. S5). These functional distinctions underscore the unique role of T14_HLA-DR^+ cells in immune responses, distinguishing them from classical APCs.

Compared to normal/adjacent tissue, T14_HLA-DR^+ cells in the TME exhibited significantly upregulated genes associated with antigen-presenting capabilities, including MHC II molecules (e.g., *HLA-DRA* and *HLA-DRB1*), MHC II stabilizer/transporter (e.g., *CD74*), and related pathways (Fig. 4D and fig. S7C). Notably, these genes were not upregulated in DCs and were even downregulated in macrophages, suggesting a potential TME-specific induction of antigen presentation in the T14_HLA-DR^+ subset (Fig. 4E). Moreover, increased cell-cell communications were observed in tumor samples between γδ T subsets (with T14_HLA-DR^+ ranking highest) and other cell types, particularly with $CD8^+$ T cells and $CD4^+$ T cells (Fig. 4F). Deconvolution analysis further revealed that patients with higher T14_HLA-DR^+ infiltration exhibited an increased proportion of $CD4^+$ and $CD8^+$ T cells in the TME across multiple cancer types (Fig. 4G and fig. S7D). Additional validation via mIF assays confirmed that T14_HLA-DR^+ closely interacts with $CD4^+$ and $CD8^+$ T cells across various cancer types, reinforcing their co-localization with these T cell subsets (Fig. 4H).

To further clarify the specific T cell subtypes engaged by T14_HLA-DR^+ antigen presentation, spatial transcriptomics-based deconvolution analysis demonstrated significant co-localization of T14_HLA-DR^+ with $CD4^+$ Tex and $CD8^+$ Tex cells (Fig. 5, A and B, and fig. S8), consistent with the results of our cell-cell communication analysis (Fig. 4A). These results supported the role of T14_HLA-DR^+ as APCs that promote the activation of specific T cell subtypes. Additionally, ligand-receptor analysis revealed the MHC II-CD4 pair between T14_HLA-DR^+ and $CD4^+$ Tex cells, which may facilitate $CD4^+$ T cell activation (Fig. 5C). We also identified the MHCII-LAG3 interaction between T14_HLA-DR^+ and $CD8^+$ Tex cells (Fig. 5C), consistent with previous reports implicating LAG3 in the negative regulation of activated T cells in mice [75, 76]. These interactions were spatially

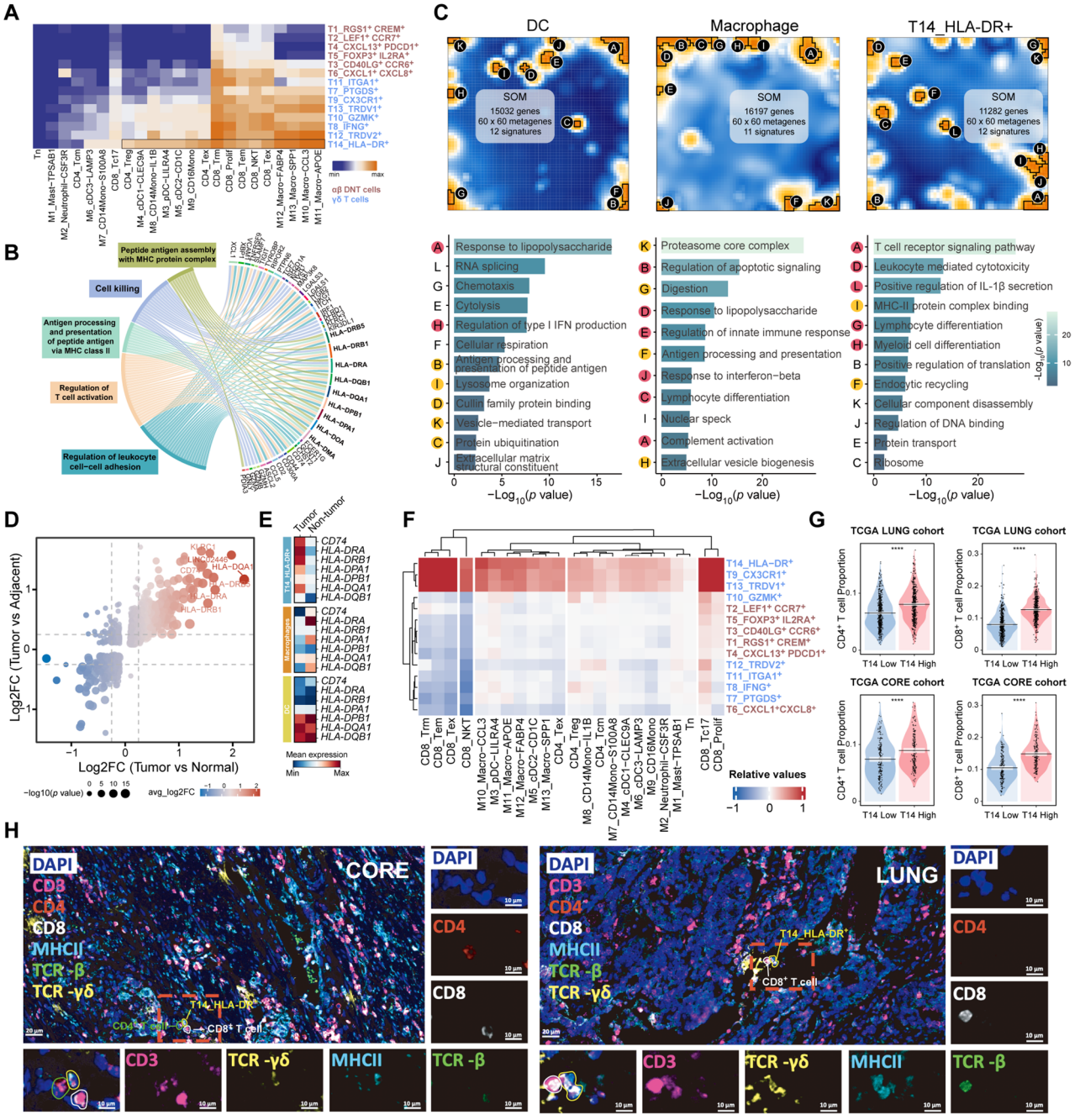


Fig. 4 Antigen presentation function of T14_HLA-DR$^+$ cells. **A** Interactions between DNT cell subsets and other cell types in the tumor microenvironment. **B** Chord diagram illustrating Differential expressed genes (DEGs) in T14_HLA-DR$^+$ cells enriched for GO pathways related to antigen presentation and cytotoxic immune responses. **C** (Top) SOM (60 × 60 grid) constructed from single-cell transcriptomes of DC (left), macrophages (middle), T14_HLA-DR$^+$ (right) cells showing scaled metagene expression. Overexpressed metagene signatures are outlined by black lines. (Bottom) Function enrichment of overexpressed metagenes analyzed by SOM. Dark to light indicates *p* values from low to high, respectively. Yellow circles indicate antigen processing and presentation pathways; red circles indicate immune response/regulation pathways. **D** DEGs of T14_HLA-DR$^+$ cells comparing tumor versus non-tumor. X-axis represents DEG between tumor and adjacent tissues, while y-axis represents DEG between tumor and normal tissues. The color and size of the points represent -log10(*p* value). **E** The average expression of antigen-presentation-related genes in tumor and non-tumor tissues for T14_HLA-DR$^+$ cells (top, blue), macrophages (middle, orange), and DCs (bottom, yellow). **F** Heatmap depicting differential interaction strength in the cell-cell communication network between tumor and non-tumor regions, with red indicating increased signaling in tumors compared to non-tumor regions. **G** Percentages of CD4$^+$ (left) and CD8$^+$ (right) T cells among all CD3$^+$ T cells in TCGA LUNG (top) and CORE (bottom) cohort, stratified by the median abundance of T14_HLA-DR$^+$ cells in these patients. The *p*-values calculated using the Wilcoxon test. **H** Multiplex immunofluorescence staining showing the co-localization of T14_HLA-DR$^+$ and CD8$^+$ or(and) CD4$^+$ T cells in CORE (left) and LUNG (right) samples. Scale bars, 10 μm. * *p* value < 0.05, ** *p* value < 0.01, *** *p* value < 0.001, and **** *p* value < 0.0001

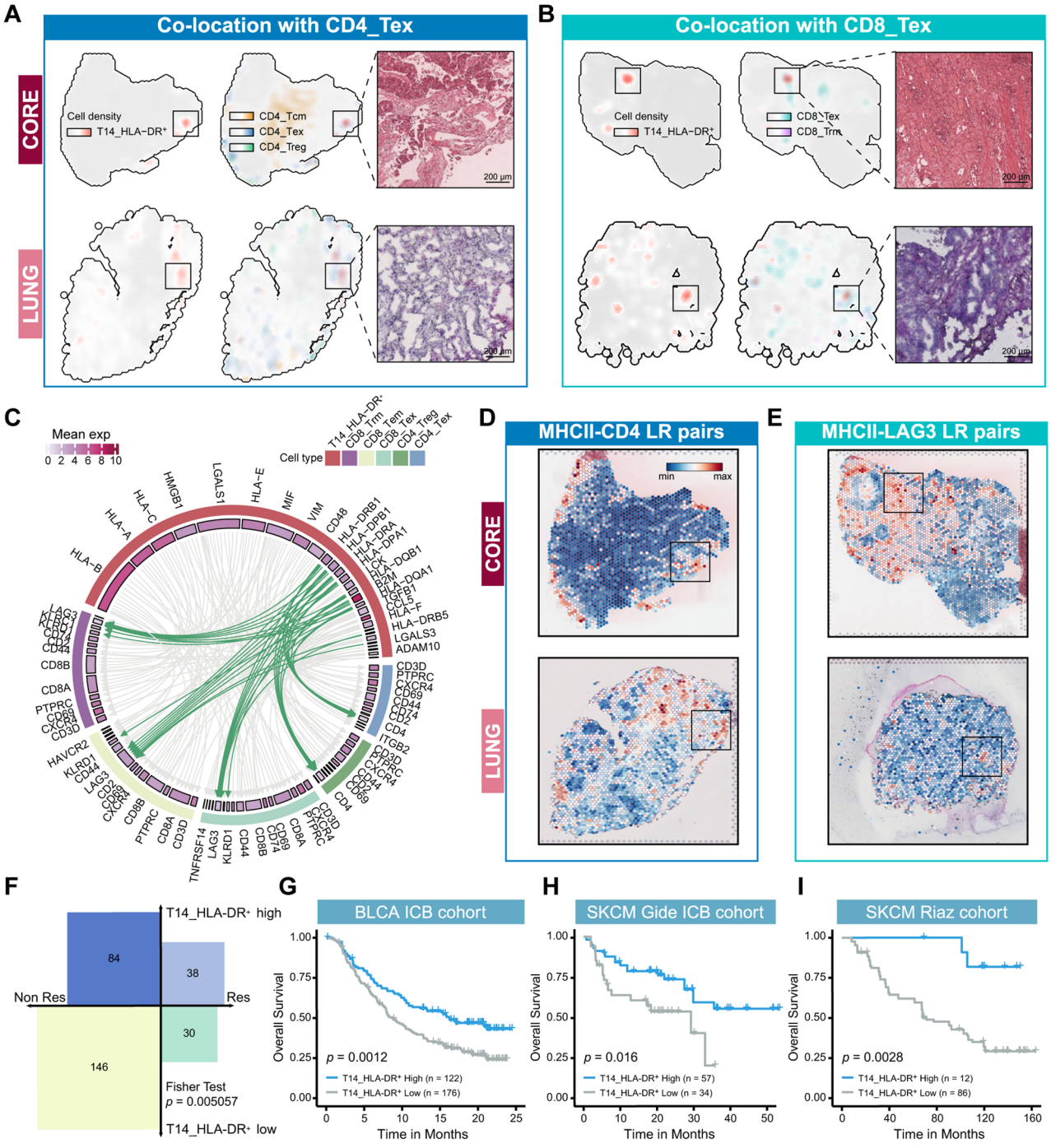
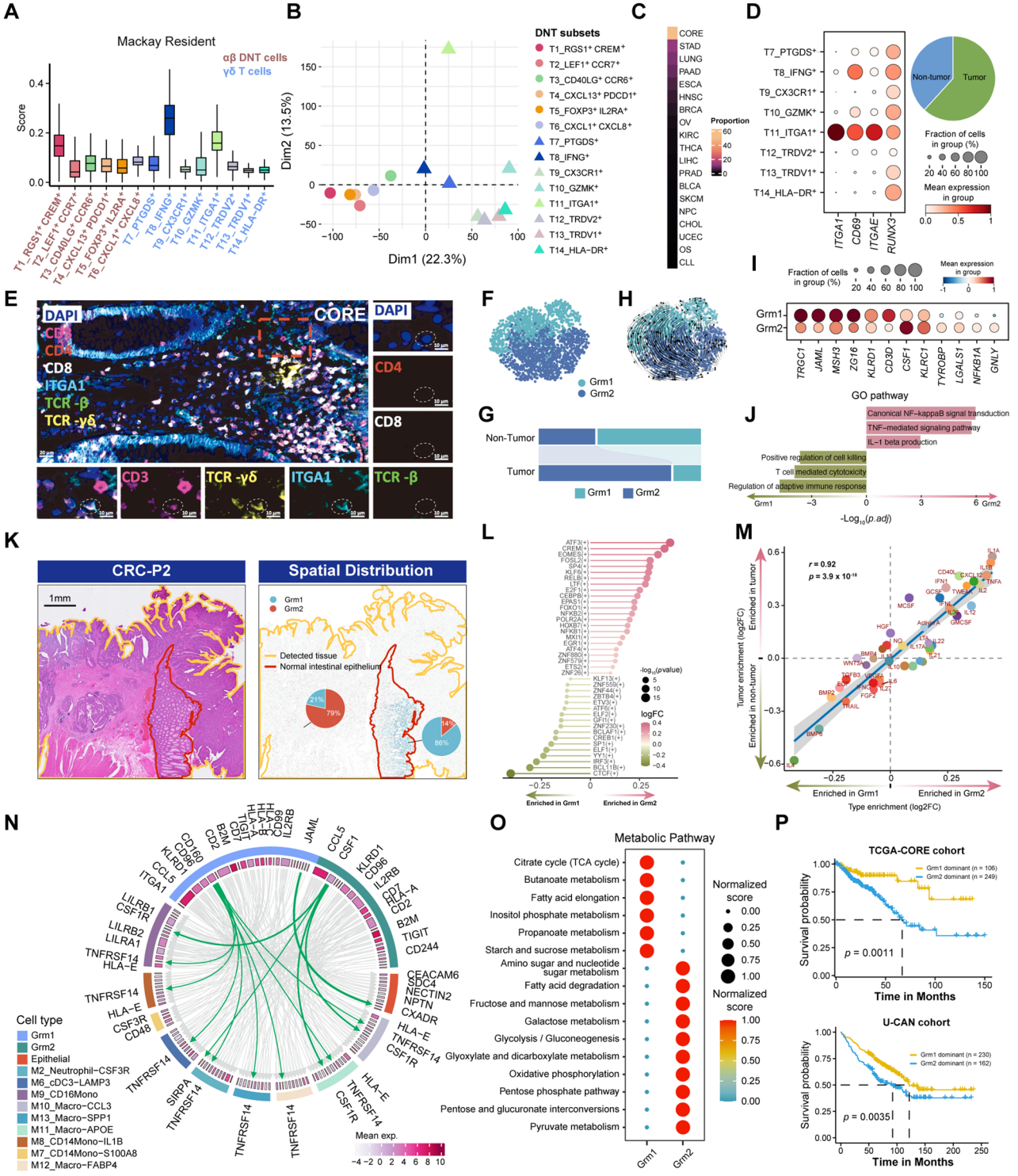


Fig. 5 Cell-cell interactions between T14_HLA-DR$^+$ cells and other T cells. **A** and **B** Co-localization between T14_HLA-DR$^+$ cells and exhausted CD4$^+$ (**A**) or CD8$^+$ (**B**) T cells in spatial transcriptomics data from CORE (top) and LUNG (bottom) cancers. Localized magnified regions are counterstained with Hematoxylin-eosin (HE). Scale bars, 200 μm. **C** Chord plot illustrating ligand-receptor pairs from consensus databases compiled by the LIANA software. Arrows point from the ligand toward the receptors, with only HLA class-II pairs highlighted. **D** and **E** Spatial mapping of MHC class-II-CD4 (**D**) and MHC class-II-LAG3 (**E**) pairs gene expression on spatial transcriptome slides from CORE (top) and LUNG (bottom) cancers. MHC class-II (HLA-DRB1, HLA-DPB1, HLA-DRA, HLA-DPA1, HLA-DQB1, HLA-DQA1) and CD4/LAG3 pairs as identified in Fig. 5**C**. **F** In the BLCA ICB cohort, patients stratified by T14_HLA-DR$^+$ abundance showed significant differences in response rates. T14_HLA-DR$^+$ high group (31.15%) had a significantly higher response rate than the low group (17.05%). The *p*-value was calculated by Fisher's exact test. **G**, (**H**) and (**I**) Kaplan-Meier overall survival curves for the BLCA (**G**), SKCM Gide (**H**) and SKCM Riaz (**I**) ICB cohorts grouped by T14_HLA-DR$^+$ cells abundance. The *p* values were calculated by log-rank test



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concentrated in regions where T14_HLA-DR$^+$ co-localized with CD4$^+$ and CD8$^+$ Tex cells (Fig. 5, D and E, and fig. S9), providing further insights into the mechanisms by which T14_HLA-DR$^+$ cells mediate antigen presentation in the TME.

Given the dual role of T14_HLA-DR$^+$ cells as tumor-enriched APCs and cytotoxic T lymphocytes in anti-tumor immune response, we further evaluated their potential impact on immunotherapy outcomes. Analysis of several immunotherapy cohorts revealed that

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Fig. 6 T11_ITGA1$^+$ subset: Gut resident γδ T cells. **A** Box plot showing the resident signatures from Mackay's research in each DNT cell subset. **B** PCA based on average gene expression of DNT cells revealed that T11_ITGA1$^+$ subset exhibits distinct transcriptional signatures compared to other subsets. **C** Heatmap showing the frequencies of T11_ITGA1$^+$ subset across various tumor types. **D** (Left) Dot plot showing the expression of gut-resident related genes in each DNT cell subset. (Right) Heatmap showing the sample frequencies of T11_ITGA1$^+$ subset. **E** Representative example of CORE samples stained by mIF to show T11_ITGA1$^+$ cell (dashed circles). Scale bars, 10 μm. **F** UMAP plot of T11_ITGA1$^+$ cells, colored by associated cell subtypes. **G** The proportion of Grm1 and Grm2 in tumor and non-tumor tissues. **H** RNA velocity streamline plot visualizing the trajectory of T11_ITGA1$^+$ cells, colored by cell types. **I** Dot plot showing the expression of marker genes in Grm1 and Grm2 cells. **J** GO enrichment pathways of DEGs between Grm1 and Grm2 cells. **K** HE-stained CRC P2 sample (left), with the spatial mapping of Grm1 (blue) and Grm2 (red) shown (right). The red line represents the normal intestinal epithelial region, and the yellow line indicates the detected boundary. Grm1 is highly expressed in the normal epithelium than Grm2, whereas the opposite is observed in the tumor tissue. Scale bars, 1 mm. **L** Lollipop plot showing the transcription factors specifically enriched for Grm1 (green) and Grm2 (pink). **M** Grm1/2 and tumor/non-tumor cytokine enrichment. Each point represents a cytokine with a unique color. The Spearman's correlation coefficient and its p value are shown. **N** Chord plot illustrating ligand-receptor pairs from consensus databases compiled by the LIANA software. Arrows point from the ligand toward the receptors, with only selected pairs highlighted. **O** The metabolism activity of Grm1 and Grm2. The dot size and color scale represent the normalized score. **P** Kaplan-Meier overall survival curves for the TCGA-CORE (top) and U-CAN (bottom) cohorts, grouped by Grm1/Grm2 ratio. The *p* values were calculated by log-rank test. TCGA-CORE: TCGA COAD and READ cohort

patients with higher T14_HLA-DR$^+$ infiltration exhibited improved response rates to checkpoint inhibitor treatments and favorable prognoses (Fig. 5F and I). These findings highlight T14_HLA-DR$^+$ cells as promising targets for enhancing the efficacy of immunotherapy.

Tissue-resident DNT subsets

A systematic comparison of tissue-resident characteristics across all DNT subsets revealed that the T8_IFNG$^+$ and T11_ITGA1$^+$ subsets exhibited strong tissue-resident features, characterized by high expression of resident marker genes [77] (Fig. 6A). Moreover, principal component analysis (PCA) of average gene expression profiles further distinguished T11_ITGA1$^+$ from other subsets, highlighting its unique tissue-resident identity (Fig. 6B). Notably, T11_ITGA1$^+$ predominantly originated from CORE samples (Fig. 6C), displayed elevated expression of key tissue-resident and memory T cell markers, including *ITGA1*, *CD69*, *ITGAE*, and *RUNX3* (Fig. 6D). Further validation using mIF assays confirmed the prominent infiltration of T11_ITGA1$^+$ cells within CORE samples (Fig. 6E and fig. S10A). These findings align with recent observations of gut-resident γδ T cells in a mouse model [25]. Collectively, the T11_ITGA1$^+$ subset represents a distinct population of gut-resident γδ T cells, reinforcing the notion that the gut serves as a major site for γδ T cell infiltration.

To further investigate the roles of T11_ITGA1$^+$ cells within the intestinal TME, we performed reclustering of the T11_ITGA1$^+$ subset, identifying two distinct functional states, designated as Grm1 and Grm2 (Fig. 6F). These states exhibited different tissue distribution patterns (Fig. 6G and fig. S10B), with pseudotime analysis revealing a transition from Grm1 to Grm2 in gut tissues (Fig. 6H). Specifically, Grm1 was predominantly localized in non-tumor intestinal tissues (Fig. 6G and fig. S10B) and expressed high levels of genes associated with intestinal homeostasis and repair (e.g., *JAML* [78]) as well as cytotoxicity (e.g., *KLRD1*) (Fig. 6I), indicating its potential role in maintaining immune homeostasis and exerting anti-tumor activity. In contrast, Grm2 was primarily found in tumor tissues and exhibited upregulation of genes related to immunosuppression and inflammation (e.g. *KLRC1* [79] and *NFKBIA* [80]) (Fig. 6, G to I and fig. S10B). Enrichment analysis of their DEGs further supported these findings, with Grm1 being associated with anti-tumor pathways, such as positive regulation of cell killing, T cell-mediated cytotoxicity, and regulation of adaptive immune responses (Fig. 6J). Conversely, Grm2 was enriched in pro-inflammatory signaling pathways, including canonical NF-κB signal transduction, TNF-mediated signaling, and IL-1β production (Fig. 6J).

To validate these findings, we utilized public data resources. High-resolution Visium HD deconvolution of FFPE CORE tumor samples [81] demonstrated Grm1 enrichment in normal intestinal epithelium, whereas Grm2 was diffusely distributed within tumor regions (Fig. 6K and fig. S10C). In an independent external CORE immunotherapy cohort [82], we examined gut-resident subsets and identified a population phenotypically resembling T11_ITGA1$^+$ cells (CD3E$^+$TRDC$^+$ITGA1$^+$ITGAE$^+$). Based on the expression of the Grm1- and Grm2-associated marker genes defined in the atlas, this population could be further stratified into two Grm1-like and Grm2-like states of gut-resident T11_ITGA1$^+$ γδ T cells (fig S10D and E). These two states showed strong concordance with Grm1/Grm2 subsets (fig S10F). A similar distribution pattern for Grm1 and Grm2 was also observed in the cohort (fig. S10G). Notably, in the complete response group, the proportion of Grm1-like cells gradually increased over the treatment course, while an opposite trend was observed in the partial response and non-response groups (fig. S10H). These findings not only corroborate our analysis but also underscore the potential role of Grm1 cells in mediating effective antitumor immunity.

We next characterized functional differences between Grm1 and Grm2 through transcriptional regulator profiling and cytokine activity analysis. The two subsets exhibited distinct transcription factor signatures (Fig.

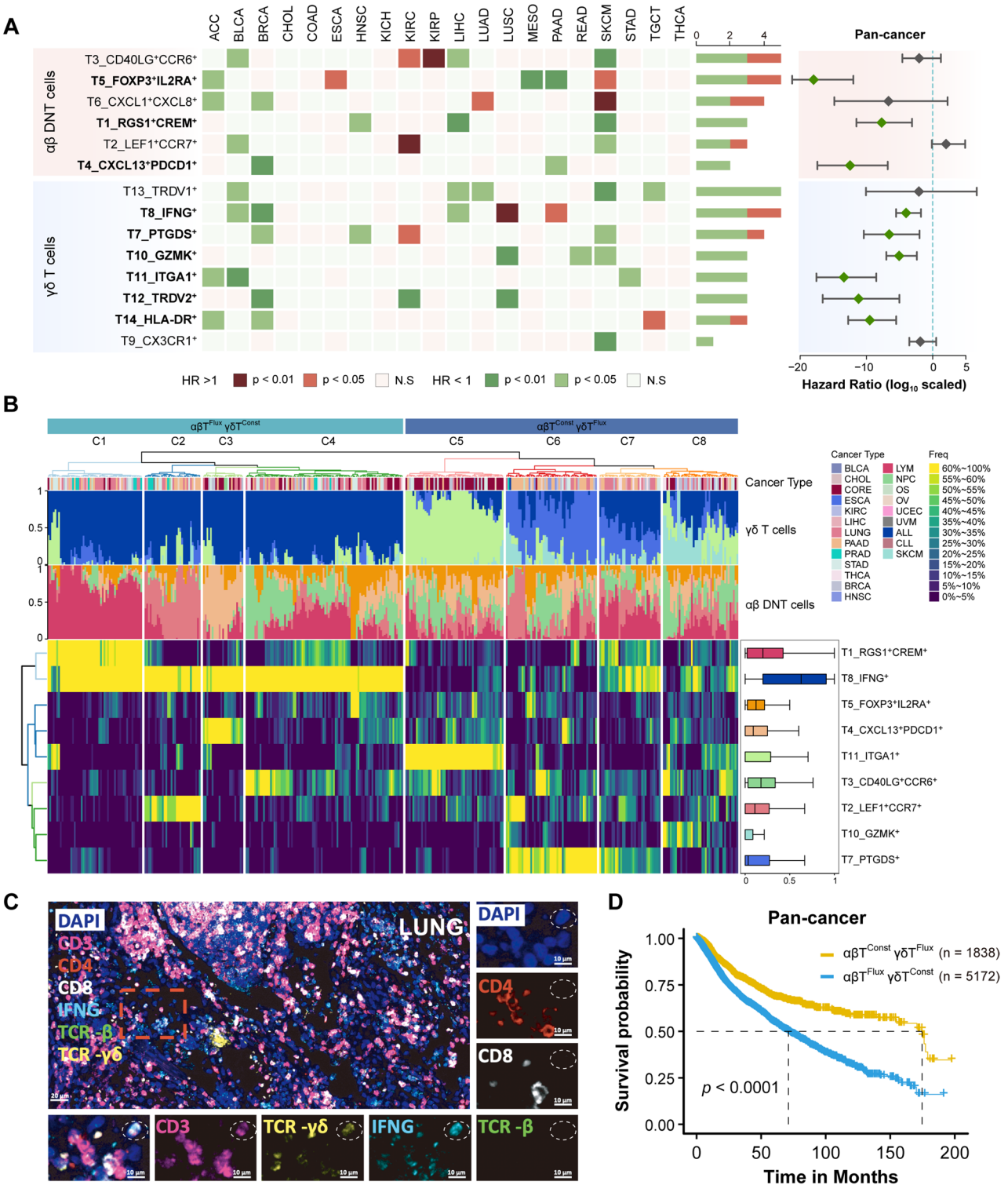

Fig. 7 (See legend on next page.)

6L), with cytokine activity patterns strongly correlating with their tissue-specific origins (Fig. 6M). Particularly, Grm2 exhibited high ATF3 activity (Fig. 6L), which has been shown to induce IL-17 A production via NLRP3 inflammasome-dependent IL-1β secretion [83]. Consistently, cytokine signaling activities predicted by CytoSig confirmed IL-1β and IL-17 A secretion in Grm2 (Fig. 6M), aligning with the established pro-tumorigenic roles

(See figure on previous page.)

Fig. 7 Impact of DNT subsets on prognosis. **A** (Left) Hazard ratios calculated by multiple Cox proportional hazards models with overall survival, adjusted forage and sex, tumor type and tumor stage, in the each TCGA cohort. Hazard ratios greater (less) than 1 are colored in red (green), with color intensity increasing as the p-value becomes significant. (Right) Forest plot depicting the hazard ratios calculated by multiple Cox proportional hazards models with overall survival, adjusted for tumor stage, age and sex, across the TCGA cohorts at pan-cancer level. Hazard ratios with 95% confidence interval are depicted (solid lines). The *p* values were calculated by Wald test. Green block represents better survival ($p < 0.05$); grey block indicates $p > 0.05$. **B** Heatmap showing the frequencies of DNT cell subsets in the tumor tissue. Each column represents an individual patient. Each row represents the frequency for each DNT cell subset. Subsets with low frequency were excluded. On the right, a box plot illustrates the distribution of DNT subset proportions across samples. Hierarchical clustering was performed using Euclidean distance as the metric. **C** Representative example of LUNG cancer sample stained by multiplex immunofluorescence to show T8_IFNG$^+$ cells (dashed circles). Scale bars, 10 μm. **D** Kaplan-Meier overall survival curves for TCGA cohorts at pan-cancer level, grouped by DNT compositions. The *p* value was calculated by log-rank test

of IL-17-producing γδ T cells [8]. Additionally, Grm2 expressed MCSF (CSF1), which may promote macrophage polarization toward the immunosuppressive and protumorigenic M2 phenotype via the CSF1-CSFR axis [84], as revealed by ligand-receptor analysis (Fig. 6N). In contrast, Grm1 secreted higher levels of anti-tumor cytokines, including TRAIL and IFNG (Fig. 6M), and exert anti-tumor effects via JAML and its ligand, coxsackie and CXADR [85] (Fig. 6N). These findings underscored the distinct roles of Grm1 and Grm2 in shaping the TME, with Grm2 fostering immunosuppression and Grm1 supporting anti-tumor immunity.

Given the association of T cell phenotypic switch with the acidic and hypoxic conditions of the TME [86, 87], we performed metabolic analysis and revealed that Grm2 exhibited higher activity in glycolysis, oxidative phosphorylation, and fatty acid degradation pathways compared to Grm1 (Fig. 6O). Indeed, Grm2 displayed stronger hypoxia signaling and elevated HIF1A activity (fig. S10I). These metabolic features highlighted that the acidic and hypoxic conditions of the TME could drive the phenotypic transition from Grm1 to Grm2. Using the bulk transcriptomic data from TCGA CORE patients, we assess the impact of this transition on survival outcomes. Although the deconvoluted proportion of T11_ITGA1$^+$ cells did not distinguish the patients in terms of overall survival, samples with dominant infiltrations of Grm1 were associated with favorable prognosis compared to those with dominant Grm2 infiltration in TCGA-CORE cohort and a recent reported independent U-CAN cohort [88] (Fig. 6P and fig. S10J). These findings underscored the critical role of the Grm1-Grm2 balance in tumor progression.

Impact of DNT subsets on prognosis

Given the association of high level of total αβ DNT and γδ T cell with favorable prognosis across various cancer types (Fig. 1L), we evaluated the impact of individual DNT subtypes. Using deconvoluted compositions of each DNT subset across all TCGA cancer types, Cox proportional hazards regression models revealed that subsets of T13_TRDV1$^+$, T10_GZMK$^+$, T11_ITGA1$^+$, T12_TRDV2$^+$, T1_RGS1$^+$_CREM$^+$ and T4_CXCL13$^+$_PDCD1$^+$ consistently acted as protective factors in multiple cancer types independent of clinical characteristics (e.g., TNM stage) (Fig. 7A and fig. S11A). In contrast, other DNT subsets (e.g., T3_CD40LG$^+$CCR6$^+$) may play dual roles, acting as either protective (in BLCA, LIHC and SKCM) or risk factors (in KIRC and KIRP), depending on the specific cancer type (Fig. 7A).

To further investigate the clinical relevance of DNT compositions at the individual level, we stratified tumor patients into two main groups: $\alpha\beta T^{flux}\gamma\delta T^{const}$ and $\alpha\beta T^{const}\gamma\delta T^{flux}$ (Fig. 7B). The $\alpha\beta T^{flux}\gamma\delta T^{const}$ group was characterized by a marked predominance of the T8_IFNG$^+$ subset within the γδ T cell compartment, aligning with its broad distribution across multiple cancer types (Fig. 2D), as confirmed by mIF (Fig. 7C and fig. S11B). In contrast, the $\alpha\beta T^{const}\gamma\delta T^{flux}$ group exhibited a similar distribution of αβ DNT subsets but showed substantial variation in γδ T cell subsets. For instance, C5 within this group was dominated by T11_ITGA1$^+$ subset, which predominantly represented CORE patients, while C6, featured by T7_PTGDS$^+$, was primary composed of PAAD patients. Stratification of TCGA patients using these classifications revealed that the $\alpha\beta T^{const}\gamma\delta T^{flux}$ group exhibited better overall survival outcomes in specific cancer types, including THCA, KIRC, SKCM and STAD, as well as at pan-cancer level (Fig. 7D, and fig. S11, C and D). These results underscored the critical role of DNT-mediated immune responses in anti-tumor immunity, highlighting the importance of further exploring the clinical implications of DNT subsets from a pan-cancer perspective.

Discussion

In this study, we constructed a comprehensive single cell atlas of human DNT cells by integrating 2,369 samples from both publicly available and in-house datasets. We first systematically analyzed the functional differences between αβ DNT and γδ T cells, with the most striking divergence observed in their roles in the immune response. While αβ DNT cells exhibit cytotoxicity, their killing capacity is notably less pronounced compared to γδ T cell subsets. This observation is consistent with previous studies on allogeneic DNT adoptive therapy, which reported potent antitumor activity involving a mixture of αβ- and γδ- DNT cells, with the latter being

the predominant subset [7, 89]. Next, our high-resolution atlas of DNT cells corroborates several reported subsets which previously identified in mouse or non-tumor models, including FOXP3$^+$ subset [12–15], PD-1$^+$ subset [53], RORC$^+$ subset [54], the naïve subset [55], and CREM$^+$ subset [56] for αβ DNT cells, as well as IFN-γ producer [16, 17], GZMK$^+$ subset [18], APC-like subset [11, 19], gut-associated subset [25], CX3CR1$^+$ subset [57], Vδ1$^+$ subset [58, 59], and Vδ2$^+$ subset [60, 61] for γδ T cells. These results validate the accuracy of our analysis and increase the reliability of our newly identified subsets, T6_CXCL1$^+$CXCL8$^+$ for αβ DNT and T7_PTGDS$^+$ for γδ T cells.

Previous studies have indicated that γδ T cells exhibit a wide range of functions, including cytotoxicity, antigen presentation, and immunoregulation [2]. However, it remains unclear whether these diverse functions are shared across different subsets or are specific to certain ones. Our findings revealed that most γδ T cell subsets demonstrate multiple functions beyond cytotoxic capabilities. For instance, T14_HLA-DR$^+$ displays both APC-like and cell-killing characteristics, which has been previously described [11, 19, 65]. Unlike conventional APCs, T14_HLA-DR$^+$ cells are specifically activated within the TME, enhancing their antigen presentation abilities and potentially augmenting the cytotoxicity of conventional CD4$^+$/CD8$^+$ T cells [8, 64]. Furthermore, we observed that γδ T cell heterogeneity exists not only between distinct subsets but also within individual subsets. T11_ITGA1$^+$ subset, which is significantly enriched in CORE, displays dual functions in both anti-tumor and pro-tumor states. This phenotypic conversion is closely related to its localization in both non-tumor and tumor regions. Consistent with our findings, recent studies have highlighted the dual role of gut-resident γδ T cells in anti-tumor immunity, including the presence of tumor-promoting subsets [90]. Moreover, we found that gut-resident γδ T cells exhibit remarkable plasticity, enabling phenotypic conversions between anti-tumor and pro-tumor states under the influence of the TME. This plasticity underscores the dynamic interaction between γδ T cells and the TME, along with the previous reports on the crosstalk of γδ T cells with CAF [91], emphasizing the importance of considering intra-tumoral heterogeneity when developing and optimizing immunotherapeutic strategies.

Clinically, the presence, subtype composition, and functional states of DNT cells may significantly influence treatment responses and patient survival outcomes, particularly in relation to cancer immunotherapy [9, 92]. Although many DNT subsets demonstrate robust anti-tumor activities [8], the lack of significant prognostic associations in certain tumors likely reflects underlying intra-tumoral heterogeneity [64]. These findings caution against one-size-fits-all therapeutic approach to targeting γδ T cells, highlighting the need to consider their complex diversity. Instead, our study provides a framework for more precise immunotyping of tumor-infiltrating DNT cells. By integrating single cell immunophenotyping into clinical decision-making [41], it is possible to develop personalized therapies that more effectively harness the immunologic potential of DNT cells, thereby optimizing clinical outcomes.

Methodologically, our study offers several advantages. First, by utilizing a non-parametric clustering strategy that does not rely on predefined markers, we are able to identify novel subsets and define their unique transcriptional signatures based on the largest number of DNT cells captured to date. Second, compared to low-throughput techniques like flow cytometry, whole-transcriptome profiling enabled the functional analysis of both previously reported and newly identified DNT subsets. Moreover, by incorporating the broader TME cell repertoire, we conducted robust intercellular interaction analyses with well-established bioinformatic analysis [93, 94] and subsequent experimental validation, gaining valuable insights into how DNT subsets interact with other immune compartments. Lastly, our pan-cancer cohorts allowed us to explore the tissue preference and tumor specificity of DNT subsets, while also revealing conserved features across different malignancies.

Several limitations of this study should be acknowledged to avoid over-interpretation. First, we lacked comprehensive TCR sequencing data, and the incomplete availability of raw data prevented us from inferring the full TCR repertoire in many samples. This limitation restricts our ability to accurately correlate TCR composition with cellular function, particularly since most γδ T cells appeared to be dominated by *TRDV1* in tumor [92, 95]. Second, the validation of certain tumor-specific subsets was not feasible due to the limited availability of corresponding tumor samples, as our study primarily focused on lung, colorectal, and gastric cancers. Third, the inherent rarity of DNT cells poses significant challenges for experimental validation of their predicted functions. Future studies should aim to isolate and functionally characterize these subsets through large-scale cell sorting or in vivo models, which are essential for confirming their roles and identifying additional therapeutic targets.

Conclusions

In summary, by delineating the heterogeneity and functional diversity of DNT cell subsets, our study provides the most comprehensive and high-resolution single-cell atlas of tumor-infiltrating DNT, which is publicly available for further investigation on DNT cells at pan-cancer level. This resource paves the way for next-generation

immunotherapies designed to effectively harness the unique capabilities of DNT cells, thereby potentially enhancing patient treatment outcomes.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Authors' contributions

Y.X., Y.L., Y.P., and H.X. designed and supervised the project. Q.H., T.Z., Z.X.R., W.M., M.X.Q., and Y.S. analyzed the data. Q.H., T.Z., X.Y.X., F.Z., Y.P., L.Z.D., H.Y.Z., C.H.H., H.X., and Q.B.X interpreted the data. H.N.C., W.H.Z., L.B.H., and Y.Z. collected the clinical samples and information. H.Y.Y., Y.L.W., M.Q.X., Y.J.H., and J.W. performed experiments and related analysis. Q.H., T.Z., draw the diagrams. Q.H., T.Z., H.Y.Y., and H.X. wrote and revised the manuscript.

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Data availability

In-house scRNA-seq matrix data that support the findings of this study are deposited in Gene expression Omnibus (GEO: accession No. GSE275648). Additionally, the integrated scRNA-seq matrix data that support the findings of this study are deposited in Zenodo (https://doi.org/10.5281/zenodo.14586220). Public scRNA-seq and spatial transcriptomics datasets used in this study are shown in **table S1** and **table S4.** The scRNA-seq data of CORE ICB validation cohort is available from GEO with an accession number of GSE236581 [82]. Expression profiles of U-CAN CORE cohorts is obtained from ArrayExpress under accession number E-MTAB-12862 [88]. The bulk gene expression data of immunotherapy cohorts were downloaded from the GEO, The European Nucleotide Archive (ENA) and The European Genome-phenome Archive (EGA) databases via the accession number GSE91061 [96], ERP105482 [97] and EGAS00001002556 [98]. The TCGA expression data and clinical information used in this study were downloaded from UCSC Xena website (https://xenabrowser.net/datapages/). Source data are provided with this paper. The bioinformatic analysis code has been uploaded into the Github (https://github.com/zhout0827/Pancancer_DNT).

Declarations

Ethics approval and consent to participate

Fresh tumor samples were obtained from colorectal cancer, gastric cancer, and lung cancer patients who underwent surgery at the Department of General Surgery, West China Hospital, from 2021 to 2024. All patients were recruited at West China Hospital and provided written informed consent. This study was approved by the Ethics Committee of West China Hospital of Sichuan University (No. 2018(280), and No. 2021(916)).

Consent for publication

All authors have reviewed the final version of the manuscript and approve it for publication.

Competing interests

The authors declare no competing interests.

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